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# SynchronAlse

Reconciling Design and Code via Semantic Graph Matching

Cursor Track | RAISE Hackathon 2026

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# THE CRISIS OF "BRITTLE CONTRACTS"

## Why Design Systems Fracture

The Design System is an intent; code is an implementation.  
"Drift" occurs through three explicit vectors:

 **Token Drift:** Hardcoding values (Hex/Pixels) instead of using semantic variables.

 **Structural Debt:** "Rogue components" that mimic the visual design but break maintainability.

 **Temporal Drift:** Figma updates that are never propagated to existing interactions.

## The Challenge of "Taste"

A standard diff tool only sees text changes. Our AI needs to have : distinguishing a poor alignment from a necessary technical layout constraint.

# SCIENTIFIC CORE: SEMANTIC TED

We frame the drift issue as computing the **Semantic Tree Edit Distance (TED)** between the design tree and the code AST.

$$\text{Cost} ( G_D , G_C ) = \min \sum_{o \in S} w ( o )$$

Where  $w(o)$  is the semantic weight of the operation (Insertion, Deletion, Renaming) evaluated by the AI.

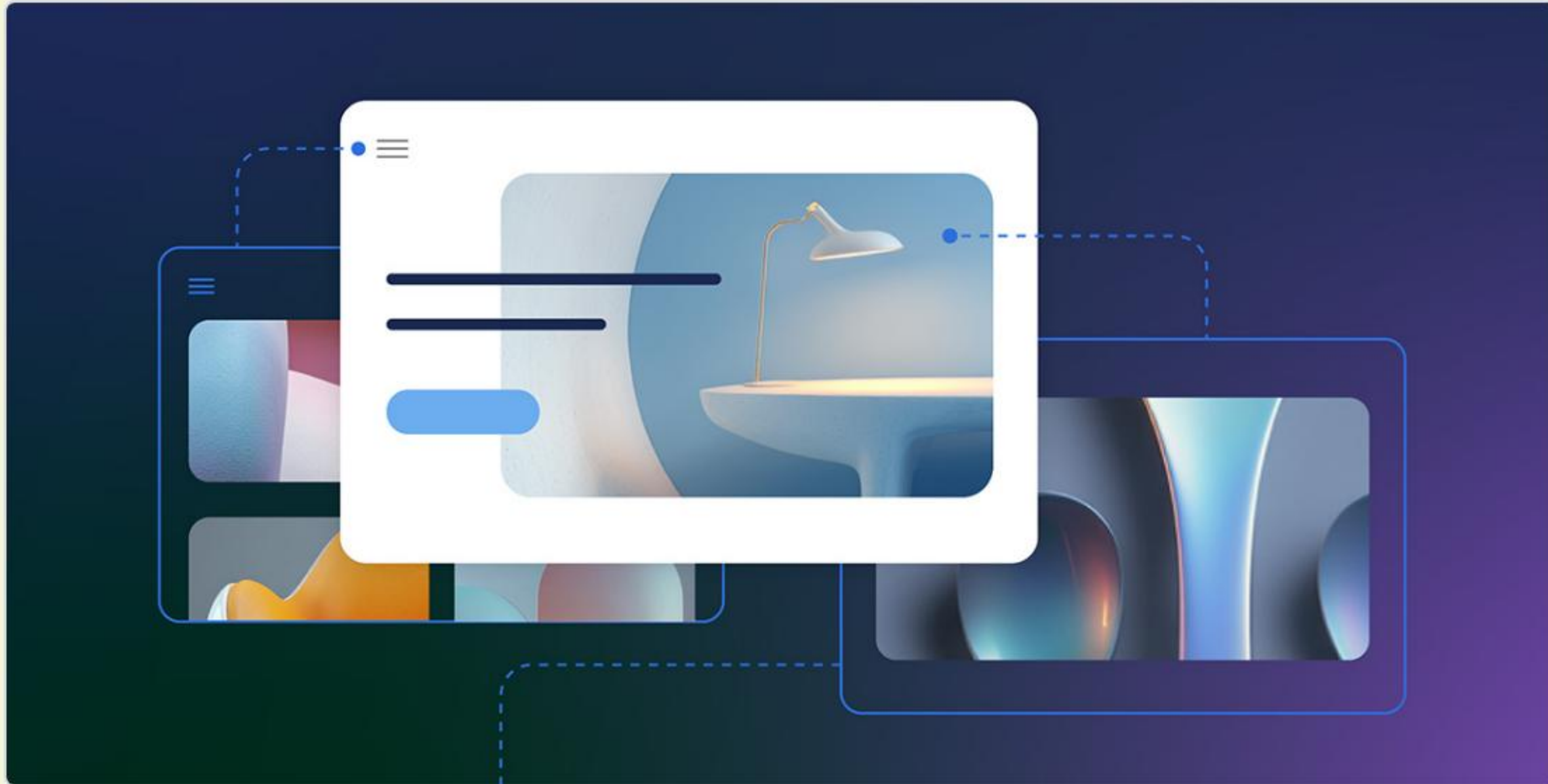
## Subgraph Matching

Identifying component instances even when the underlying DOM structure is deeper or more complex than the Figma structure.

## Token Latent Space

Projecting colors and spacing into a vector space to match "intentions" (e.g., mapping #FF0000 to token.danger).

# ARCHITECTURE: VISION-TO-SEMANTIC



## Multi-Source Extraction

1. **Figma API (Structural):** Extracting nodes, Auto-layouts, and original component IDs.
2. **Gemini 1.5 Pro (Visual):** Multimodal vision to reconstruct the observed graph directly from production pixels.
3. **Reconciliation Engine:** The agent compares both graphs and classifies anomalies by semantic severity.

# THE "TASTE ENGINE": CLASSIFICATION



## Accidental Regression

**Example:** A 15px padding instead of 16px.

**Action:** Critical alert and token suggestion.



## Technical Constraint

**Example:** Addition of a "wrapper" div for scroll management.

**Action:** Ignored (the system understands the structural necessity).



## Intentional Evolution

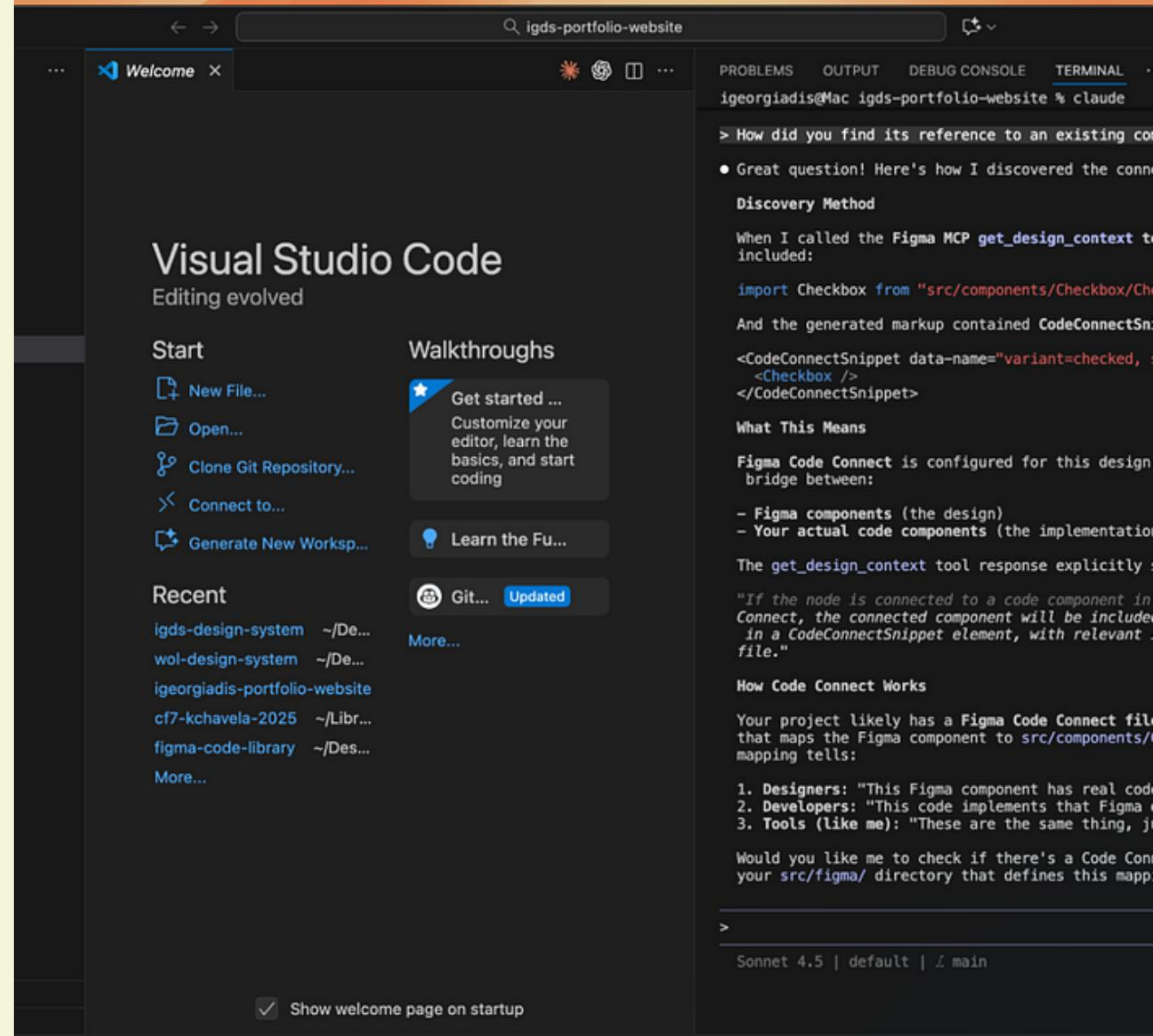
**Example:** A brand new button not present in Figma.

**Action:** Prompts for reconciliation or creation of a new official component.

# THE VISUAL PR REVIEWER

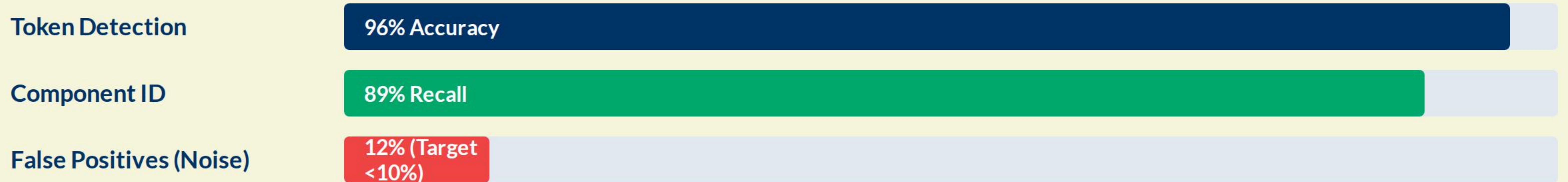
Direct integration into the CI/CD workflow. On every Pull Request, Synchronise:

- Generates a visual UI preview.
- Compares the "Snapshot" to the Figma baseline.
- Comments on the PR with the **Semantic Diff**.
- Provides the exact codebase fix to resolve the drift.



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# SEMANTIC MATCHING PERFORMANCE



*Utilizing VLMs drastically reduces false positives caused by browser rendering specificities (font rendering, subpixel anti-aliasing).*

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# CURSOR: THE CATALYST

7.4x

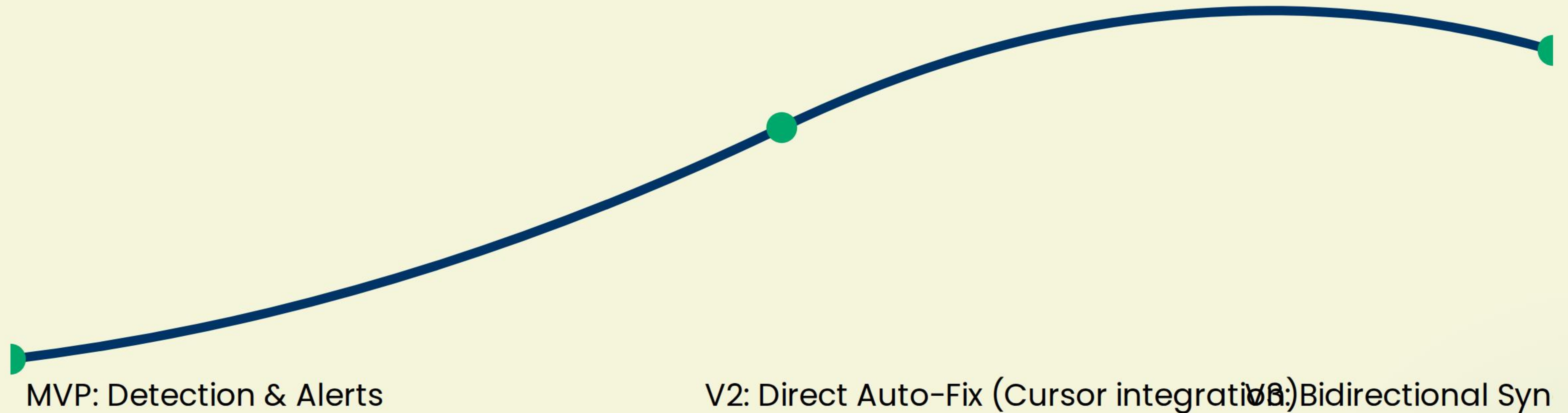
DEV ACCELERATION

## Native IDE Exploitation

Cursor was a load-bearing primitive in our process:

- **Prompt Engineering:** Rapid iteration on multimodal reconciliation prompts via @Gemini.
- **AST Refactoring:** Using Cursor to transform AI suggestions into instant multi-file code modifications.
- **Contextual Semantics:** The IDE understands our token structure, streamlining the injection of fixes.

# ROADMAP: ZERO-DRIFT ECOSYSTEM



The ultimate goal is **bidirectional synchronization**: modifying the code updates Figma, and vice-versa, with zero human friction.

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# Any Questions?

SynchronAlse: The AI that makes your Design System bulletproof.

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# IMAGE SOURCES



<https://static-academy.siteground.com/wp-content/uploads/sites/2/2025/10/Abstract-design-of-website-structure.jpg>

Source: [www.siteground.com](https://www.siteground.com)



[https://miro.medium.com/v2/resize:fit:1400/1\\*tywrih93yINEPSij0Aaag.png](https://miro.medium.com/v2/resize:fit:1400/1*tywrih93yINEPSij0Aaag.png)

Source: [uxdesign.cc](https://uxdesign.cc)